

LESSON 6

Data Gathering,
Data Preprocessing,
Data Harmonization
for intelligent system learning





(brief) EXAM SIMULATION



To TEST your preparation
(do not use the simulation to study)

Please remember



Multiple-choice test:

- 1 single correct answer $\rightarrow$ weight = 1
- N wrong answers $\rightarrow$ weight = 0
- No negative weights

1. No penalties for not answering
 $\rightarrow$ OK BUT IT'S BETTER TO TRY TO ANSWER
2. **DO NOT LOOK FOR THE RIGHT ANSWER, EXCLUDE THE WRONG ONES**
3. DOUBTS? $\rightarrow$ Choose the answer that "looks less wrong" to you

Q1



According to class discussion the theory of intelligent systems should include the following designing steps

1. *Representation*
2. *Representation, Evaluation*
3. *Representation, Evaluation, Optimization*
4. *None of the other option*

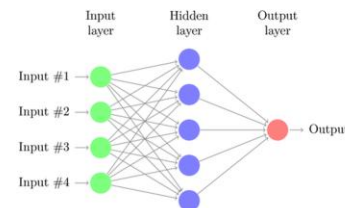
Q1



According to class discussion the theory of intelligent systems should include the following designing steps

1. *Representation*
2. *Representation, Evaluation*
3. *Representation, Evaluation, Optimization* ← correct
4. *None of the other option*

- Every machine learning algorithm has **3 components**:
 - Representation
 - Evaluation
 - Optimization



Q2



Clustering always requires a supervised dataset

1. Yes
2. No

Q2

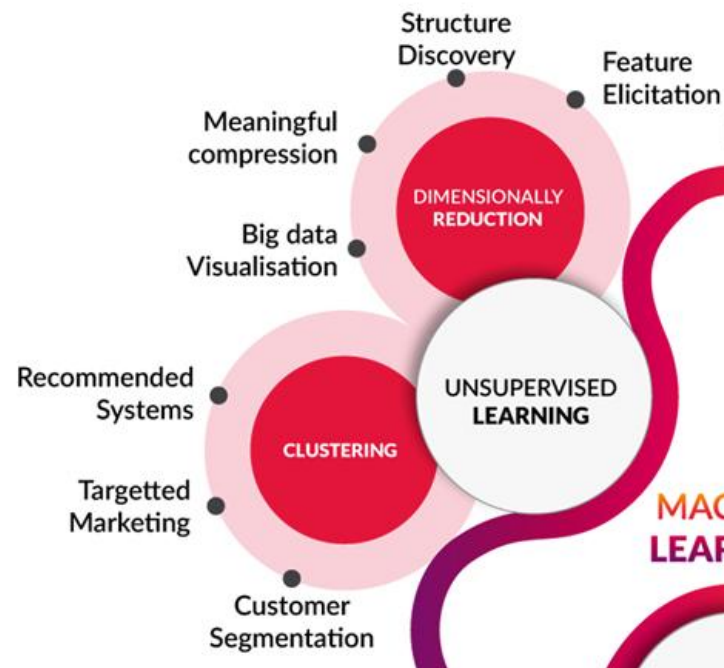


Clustering always requires a supervised dataset

1. Yes
2. No ← *Correct*

Understanding if a technique requires labels or not is a key part!

You have to master the partition of AI methods presented in Lesson 3



Q3



According to the class discussion, using a black box solution is

1. *Bad practice for a ML designer*
2. *Can be used under specific circumstances*
3. *Since all state of the art models tend to be quite large and unexplainable, it is current good practice to adopt the black box approach since you get the best models*

Q3

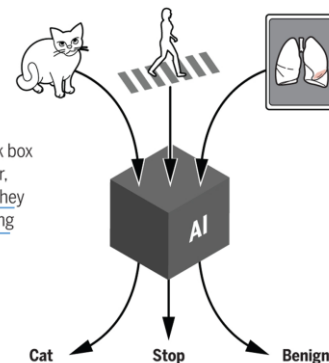


According to the class discussion, using a black box solution is

1. *Bad practice for a ML designer*
2. *Can be used under specific circumstances ← Correct*
3. *Since all state of the art models tend to be quite large and unexplainable, it is current good practice to adopt the black box approach since you get the best models*

Can I use a black box?
Yes, IF...

By definition, humans cannot assess how a black box algorithm arrives at a particular answer. However, black box methods can still provide value when they produce the best results, when the cost of a wrong answer is low, or when they inspire new ideas.



Q4



According to the class discussion to the classification systems and their decision boundaries, it is possible in general to optimize during the training/optimization step

1. *The accuracy*
2. *The margin*
3. *Both*

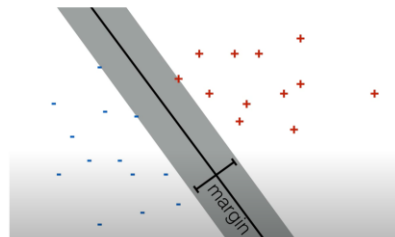
Q4



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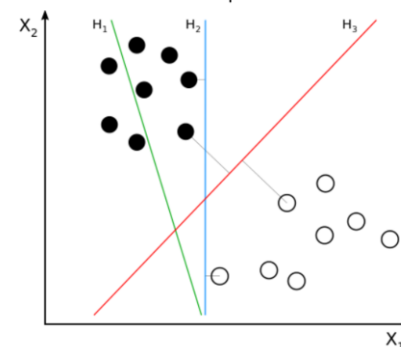
1. *The accuracy*
2. *The margin*
3. ***Both ← Correct (if the learning method you choose allow for it)***

Accuracy VS Margin



The decision boundaries H2 and H3 have both 100% accuracy (error = 0), but the margin values are different
We want to augment the margin!

We prefer H3



Q5



According to the class discussion about AI regulation in EU, the regulation approach is based

1. *List of use cases*
2. *Risk assessment of the application*
3. *Both*
4. *Non of the above*

Q5



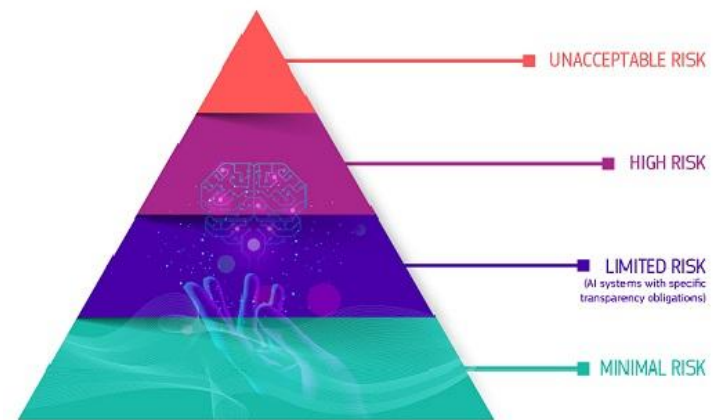
According to the class discussion about AI regulation in EU, the regulation approach is based

1. *List of use cases*
2. *Risk assessment of the application*
3. **Both ← Correct**
4. *Non of the above*

High risk

AI systems identified as high-risk include AI technology used in:

- critical infrastructures (e.g. transport), that could put the life and health of citizens at risk;
- educational or vocational training, that may determine the access to education and professional course of someone's life (e.g. scoring of exams);
- safety components of products (e.g. AI application in robot-assisted surgery);
- employment, management of workers and access to self-employment (e.g. CV-sorting software for recruitment procedures);
- essential private and public services (e.g. credit scoring denying citizens opportunity to obtain a loan);
- law enforcement that may interfere with people's fundamental rights (e.g. evaluation of the reliability of evidence);
- migration, asylum and border control management (e.g. verification of authenticity of travel documents);
- administration of justice and democratic processes (e.g. applying the law to a concrete set of facts).



Outline



Data Gathering

Where the data is coming?
How much data is coming?



Data Preparation

Data control,
Unstructured data
Data harmonization
Data analysis
Data filtering



Main points

LLM and agents



Open class discussion about data analysis

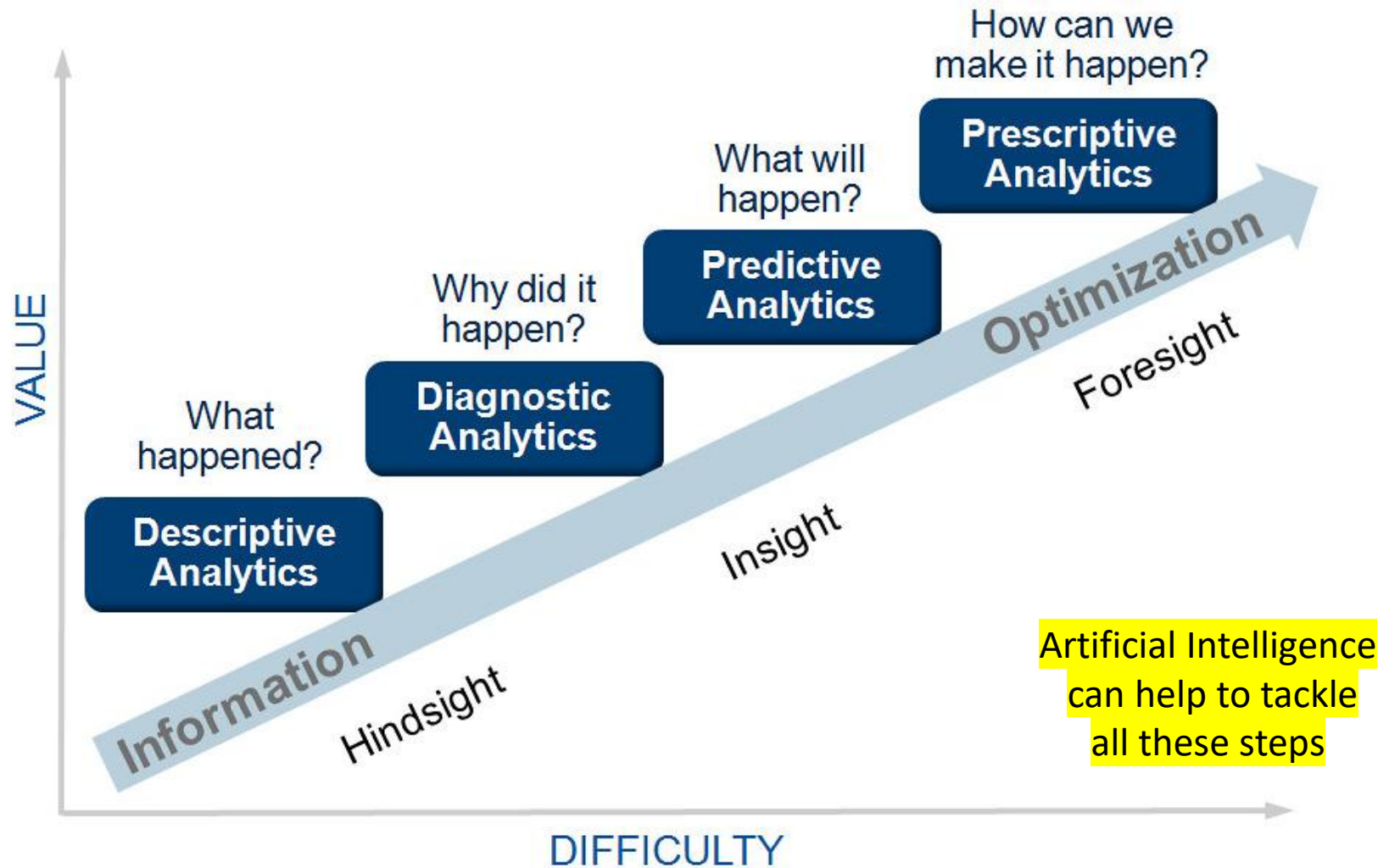
- What are your data analysis skills?
- How would you rate your data analysis skills?
- What are your previous experience?
- What are your favorite tools?
- Are you using data analysis tools for
 - Pre-processing?
 - Post-processing?
 - Results understanding?

Please, stop 1 minute
and take a note on a paper!

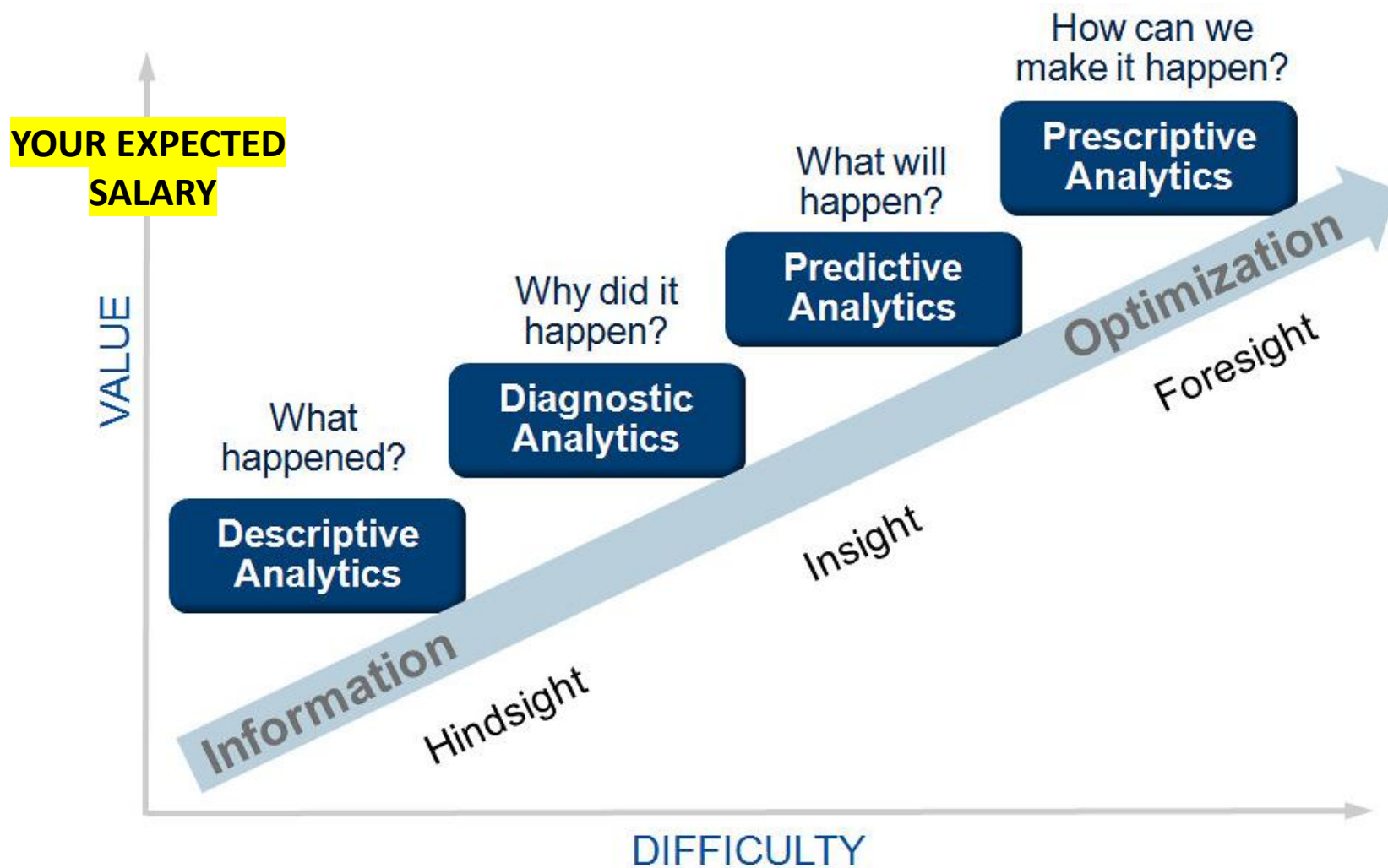
You can review your notes at the end of the course



Value of Data Analytic



Value of Data Analytic... For you

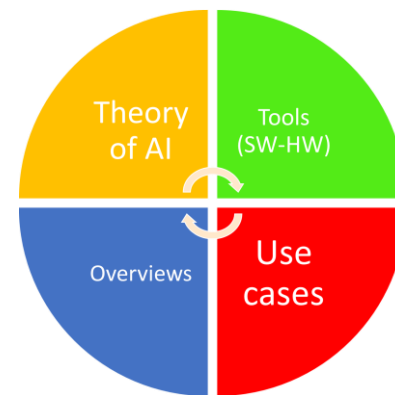




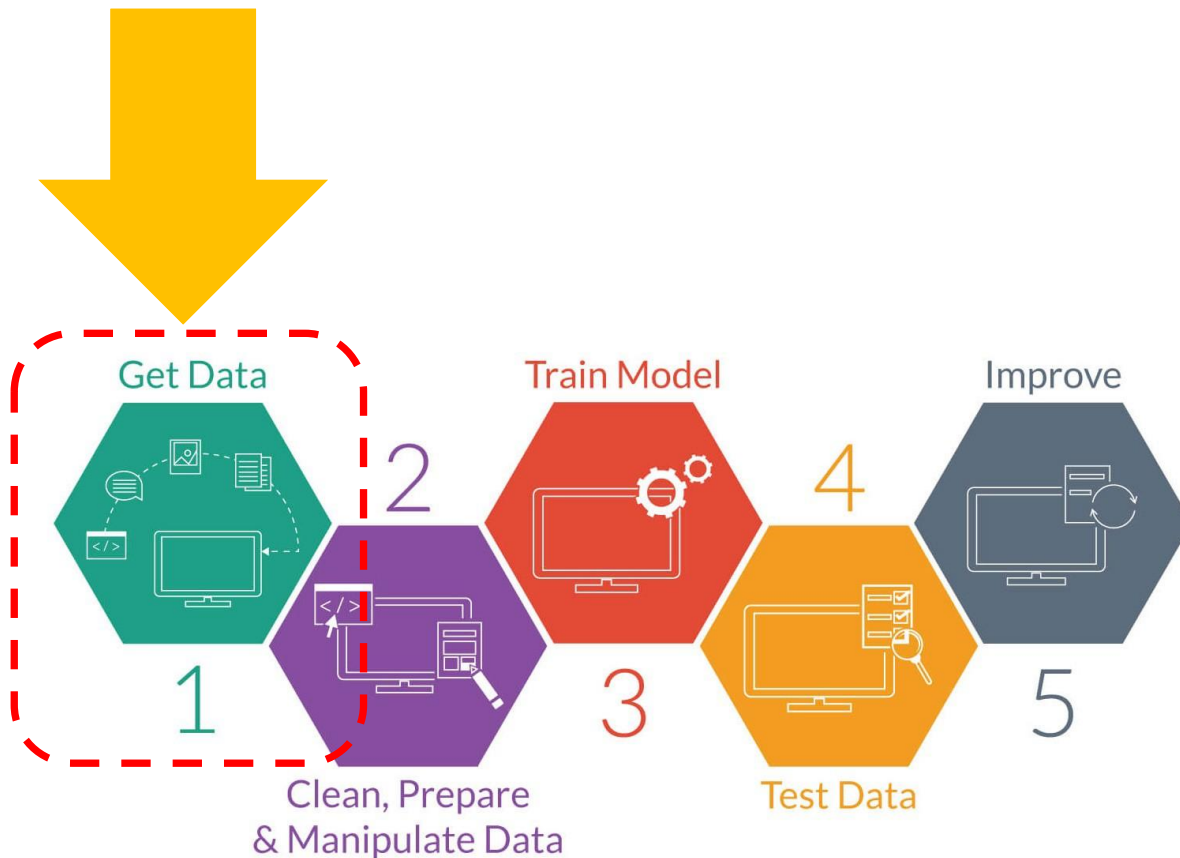
THEORY

Data gathering

Data collection and gathering:
important points



Step 1) of the ML workflow

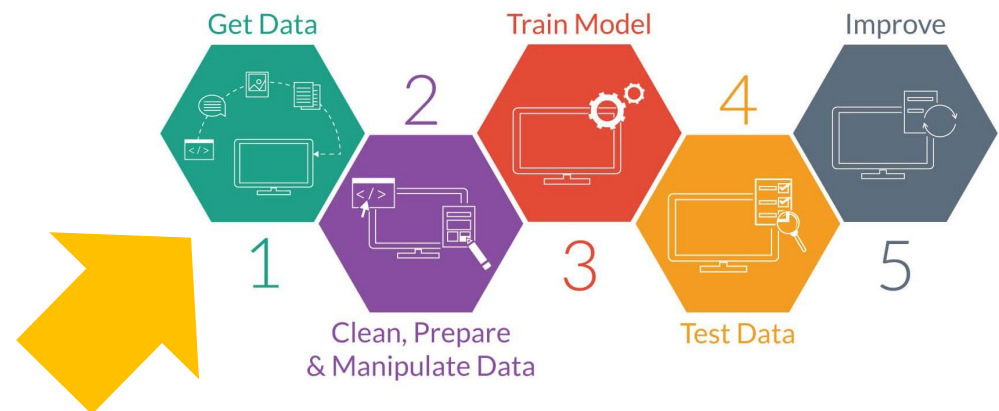


Step 1) in practice

- Classical DB query
- File processing
- Custom formats
- Stream of data from IoT and AIoT
- Online platform (AWS, Azure, Google, ...)

Outside the focus
of this course

We have so many
powerful sources
capable to generate data





Data collection

- Data collection is the process of gathering and measuring information on targeted variables in an established system
- Used in
 - Physical sciences
 - Industrial sector
 - Business
 - Humanities
 - Social sciences
 - In general, disciplines that study aspects of human society and culture
- AI is used in all this sectors
 - (even in highly unstructured data like just raw texts)



GIGO

NO «MAGICAL» NEURAL NETWORK
WILL SAVE YOU FROM THIS!!!



IoT started to generate data...



- The amount of data generated by connected internet of things (IoT) devices, in 2025 (*)
 - 41.6 billion connected devices
 - 79.4 zettabytes (ZB) of data/year.
- Trending
 - Surveillance,
 - Smart home data,
 - Smart phones
 - Coordinates, health sensors, activities
 - Wearable
 - Smart watches, bands
 - Sensors,
 - ...

Notes:

- Zettabytes = **ZB**
 → zetta indicates multiplication by the seventh power of 1000

Decimal		
Value		Metric
1000	kB	kilobyte
1000 ²	MB	megabyte
1000 ³	GB	gigabyte
1000 ⁴	TB	terabyte
1000 ⁵	PB	petabyte
1000 ⁶	EB	exabyte
1000 ⁷	ZB	zettabyte
1000 ⁸	YB	yottabyte



IoT started to generate data... (2)

- Video surveillance will drive a large share of the IoT data created
- 5G enable devices

Example:

Basic GPS coordinates from smartphones @ITA

$$\text{DataFromPositions/y} = \text{ItalianPopulation} \times \text{CellPhoneRatio} \times 365 \text{ days} \times 24\text{h/d} \times 60 \text{ min/h} \times 2\text{coordinates/min}$$
$$= 65 \times 10^6 \times 0,83 \times 365 \times 24 \times 60 \times 2 \times \mathbf{8\text{byte}} = 697996800 \text{ byte} \approx 454 \text{ TB}$$

$\sim 14,4 \text{ MB/s}$

GPS data into 48-64 bits minimum (GPS coordinates in ISO 6709 format)

<https://developers.google.com/protocol-buffers/docs/encoding>

Big Data.... Mining is needed



**BIG DATA, DATA MINING
AND CLOUD COMPUTING**



Public datasets (Annotated)



Primary source for training. Collections of Datasets!

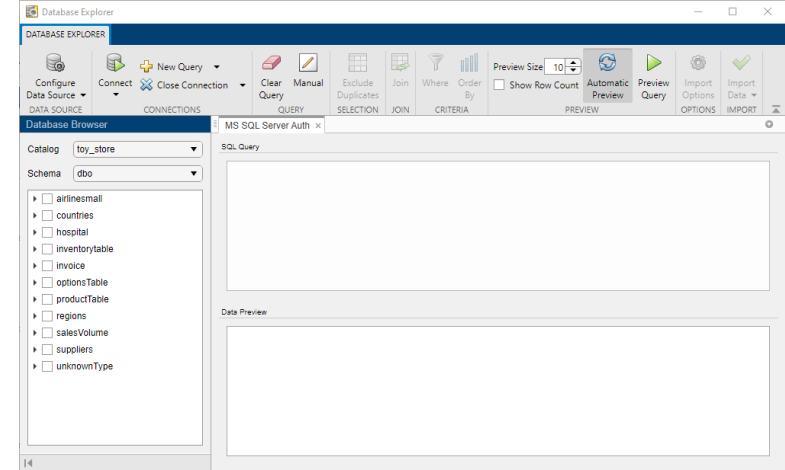
- **UCI Machine Learning Repository**
- **Kaggle Datasets**
- **Amazon Datasets**
- **Google's Datasets Search Engine**
- **Microsoft Datasets**
- **Government Datasets**
 - [EU Open Data Portal](#), [US Gov Data](#), [New Zealand's Government Dataset](#), [Indian Government Dataset](#), ...
- **Computer Vision Datasets**
- **Lionbridge AI Datasets**
- ... the list is growing on a daily basis

Tools: Matlab Database Explorer

Quickly connect to a database, explore the database data, and import data into the MATLAB workspace in a visual way

Some interesting features:

- Create and configure ODBC and JDBC data sources.
- Establish multiple connections to the same or different databases.
- Select tables and columns of interest.
- Fine-tune selections using SQL query criteria.
- Preview selected data.
- Customize import options.
- Import selected data into the MATLAB workspace for analysis.
- Save generated SQL queries.
- Generate MATLAB code.

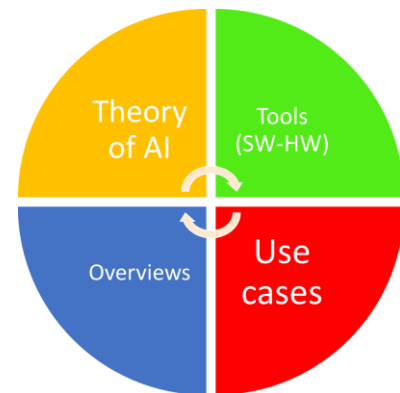




THEORY

Data heterogeneity and synchronization

Units, formats and exact timing



Data collection: data format heterogeneity

- Formats
- Name spaces
- Units
- Ranges
- Sizes
- ...

Note: Heterogeneity in statistics means that your populations, samples or results are **different**. It is the opposite of [homogeneity](#), which means that the population/data/results are the same.



Data format heterogeneity (2)

USE CASE

When NASA Lost a Spacecraft Due to a Metric Math Mistake

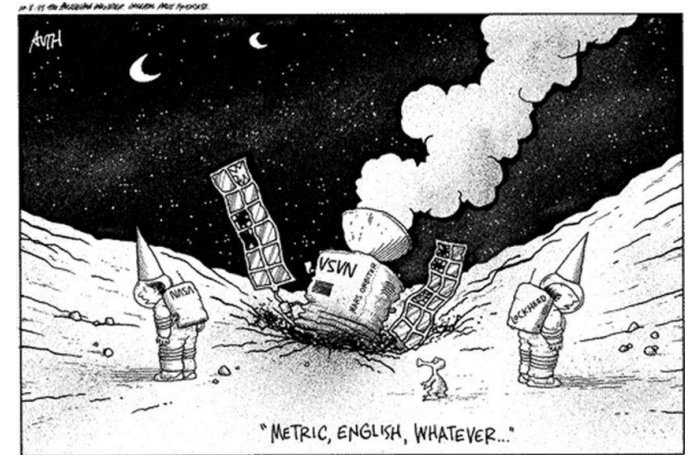
In September of 1999, after almost 10 months of travel to Mars, the Mars Climate Orbiter burned and broke into pieces. On a day when NASA engineers were expecting to celebrate, the ground reality turned out to be completely different, all because someone failed to use the right units, i.e., the metric units!

Mars Climate Orbiter was destroyed in the atmosphere...

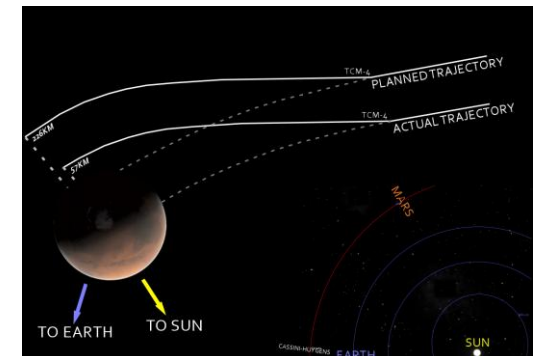
...pound-force*seconds instead of the SI units of newton*seconds....

... The Mars Climate Orbiter, built at a total cost of \$325 millions...

WRITTEN BY Ajay Harish



Remember the Mars Climate Orbiter incident from 1999?



Use case: wind power prediction

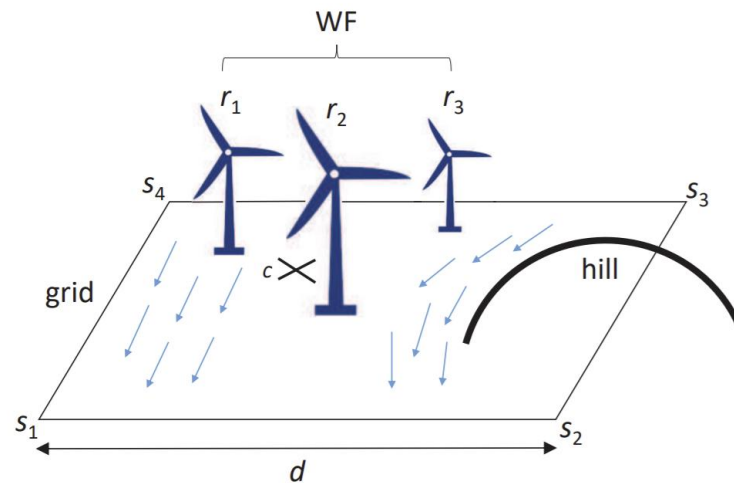
Morning/afternoon breeze



Hills/occlusions



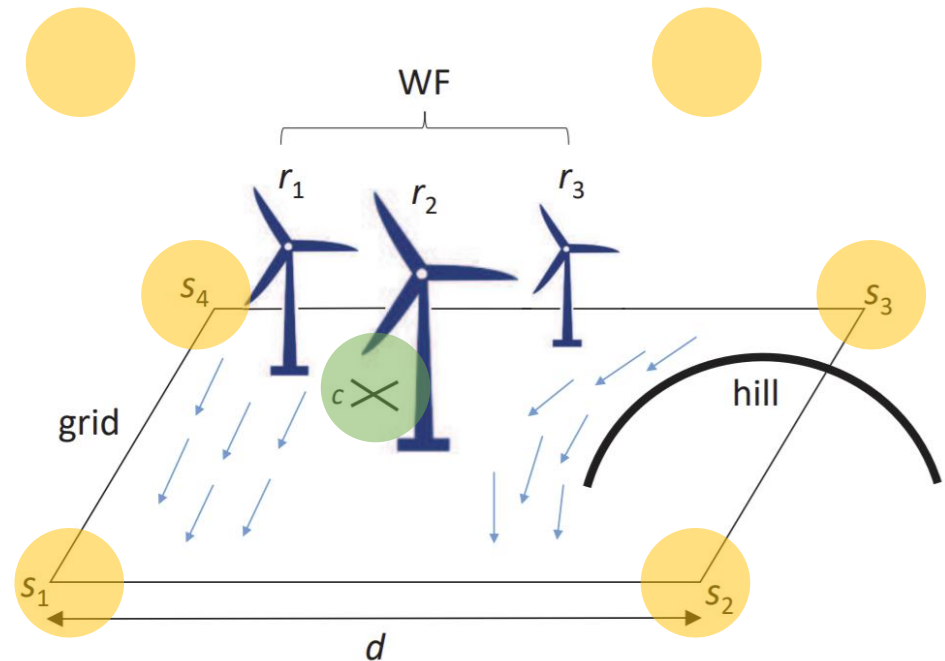
How to map the specificity
of the specific site?



Use case: wind power prediction (2)

Weather data grid has a
different altitude with
respect to the plan

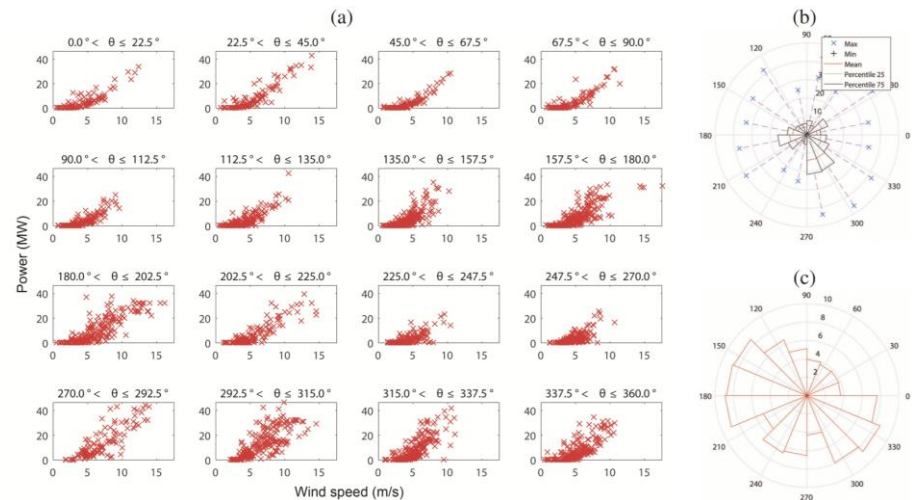
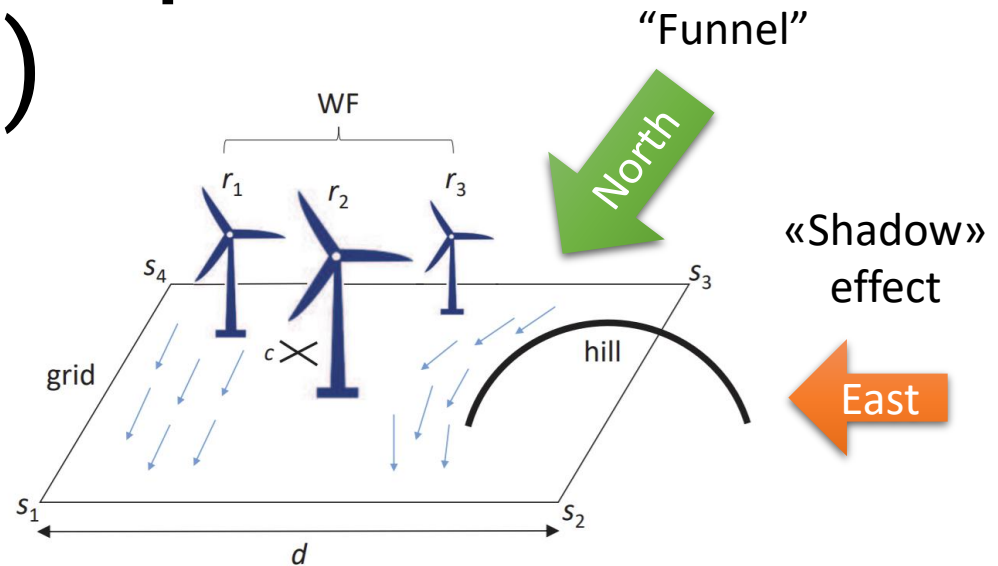
- The weather forecast data provided by different vendors and data collected by wind farm sensors could be expressed using different measurement units and reference systems
- Forecast data is different from site instrument station data (different locations)



Use case: wind power prediction (3)

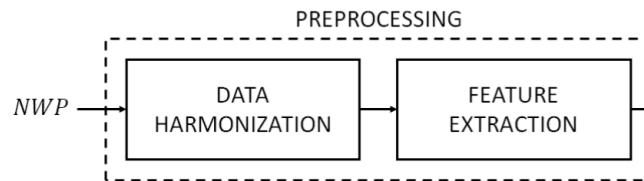
Many non idealities are present.

- Hills and occlusions can change of efficiency of the plant,
Ex.: from East $\rightarrow$ very low power
- Wind turbines has minimum speed limit to start spinning (clipping)
- Same turbines have different speed/power curves due to the surroundings condition



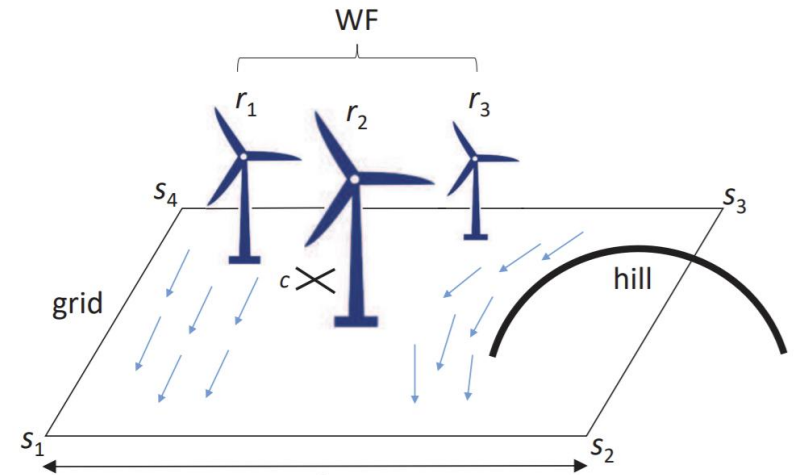
Use case: wind power prediction (4)

Goal: creation of 5y dataset from different sources of data of weather forecast data format, measurement units and reference systems

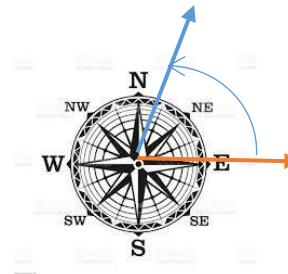
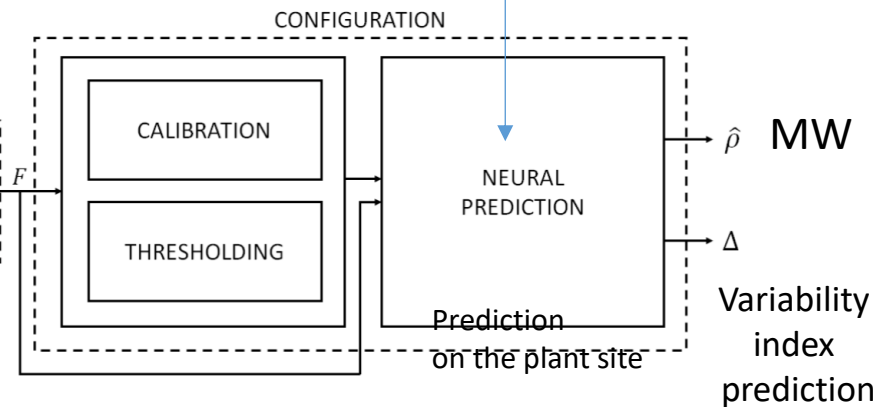


Data harmonization operation:

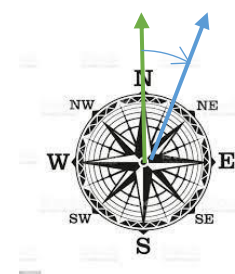
- °C for the temperature,
- m/s for the wind speed
- 0-360° angle northward directions (northward = 0°)
- hPa for atmospheric pressure,
- Scale from 0 to 1 for cloud coverage



d Weather data on the grid and different altitude



Wind angle def.1



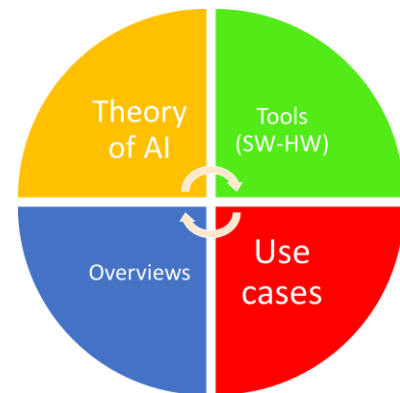
Wind angle def.2



THEORY

Data Synchronization

Keeping data in “sync”!



Data Synchronization



The way a device adjusts its internal clock in order to align with the clocks of other devices in a network

In industrial and scientific applications this part is essential

Network Time Synchronization

- Computer clocks in servers, workstations and network devices are inherently not enough accurate
 - Clocks are set by hand to within a minute or two of actual time and are rarely checked after that
 - Clocks are maintained by a battery-backed device that may drift as much as a second per day

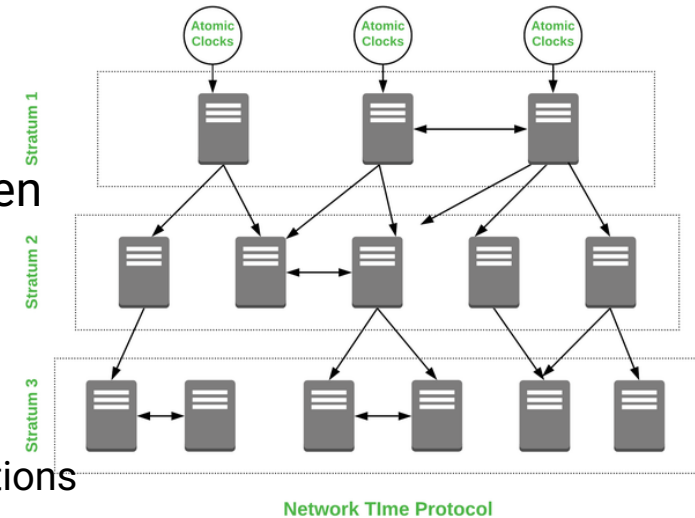
→ It's impossible to have accurate time synchronization without a proper method

NTP and Time Server

Solution 1:

Network Time Protocol (NTP) getting time from a public Internet Time Server

- NTP is a protocol for clock synchronization between computer systems over packet-switched, variable-latency data networks designed to **mitigate local network latency**
- NTP accuracy is about
 - 10s of ms over the public Internet,
 - 1ms accuracy in local area networks under ideal conditions
- Open in the firewall, UDP port 123



Solution 2:

Dedicated network **Time Server** behind your firewall

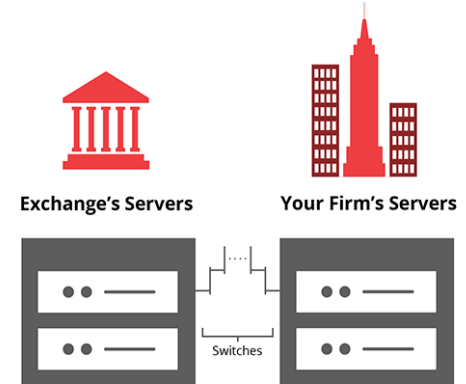
(devices synchronized to within 1/2 to 2 ms).

Solution 3: adopt the Precision Time Protocol
(standard IEEE 1588)

Network Time Synchronization Devices



Ex.: Stock Exchange



- Synchronization is key to manage the order lists, pricing and general data management

Volumes

- Bloomberg Ticker Plant: ~200 B messages/day
- OPRA (US options SIP): capacity ~1 trillion/day (post-Pillar, 2024)
- Microburst peaks: up to 187 M messages/second

Ultra-Low Latency Trading

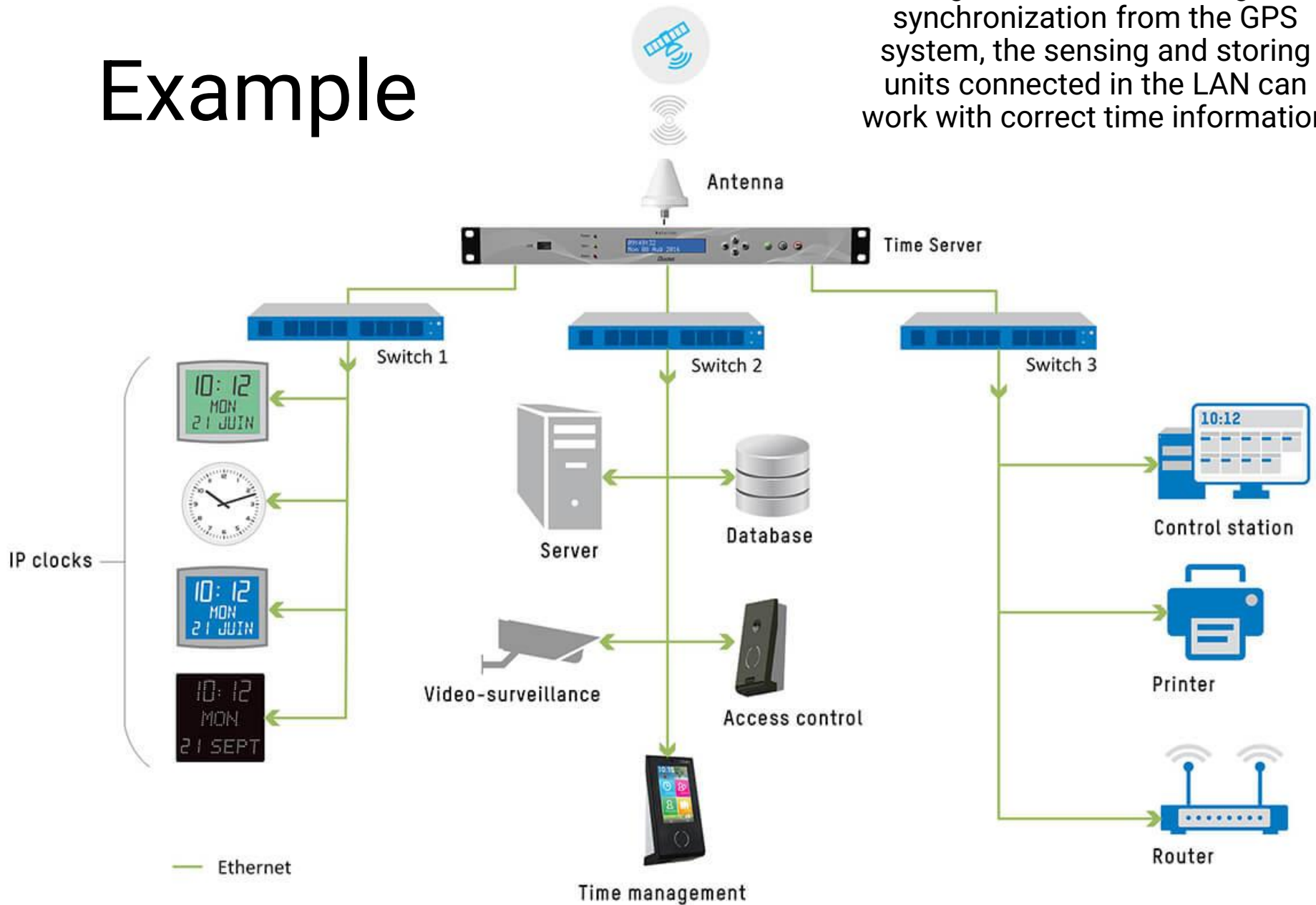
- OPRA median latency: ~20 μ s (was ~540 μ s pre-2024)
- Tick-to-trade on FPGA: 100–500 ns
- 50 μ s delay $\rightarrow$ lost queue position, worse fills
- ~70% of US equity volume is AI-driven algo trading

Time sync: NTP is no longer enough
 $\rightarrow$ Precision Time Protocol (standard IEEE 1588)

- NTP (~1 ms) is 1000 $\times$ too coarse
- Finance uses PTP (IEEE 1588) + HW timestamping $\rightarrow$ sub- μ s / ns

Example

Using a Time Server taking the synchronization from the GPS system, the sensing and storing units connected in the LAN can work with correct time information

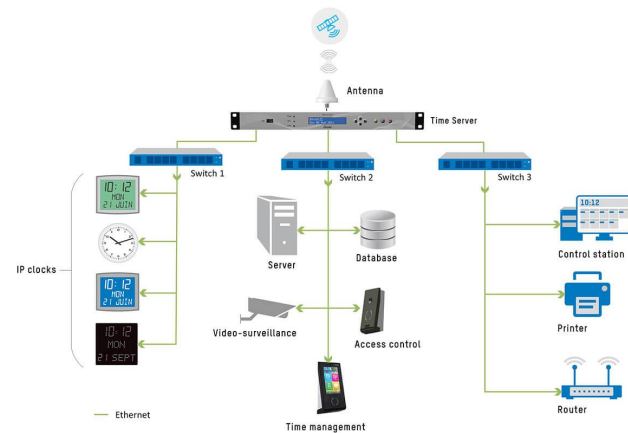


Example

According to the specific behavior of the connected sensing units using you can get data at different timestamps and missing data can occurs

You solved the synchronization, but not the sparsity of the data in the tuples

TimeStamp	T1 (K)	T2 (K)	Speed1 (m/s)	Speed2 (m/s)	RPM1 (RPM)
2026-04-03 14:27:55.021	131.5	191.4	14.5	25.2	2400
2026-04-03 14:27:58.531	131.5	191.3	NaN	NaN	2401
2026-04-03 14:28:02.343	NaN	191.3	14.5	NaN	2403



they only produce data when asked to do so

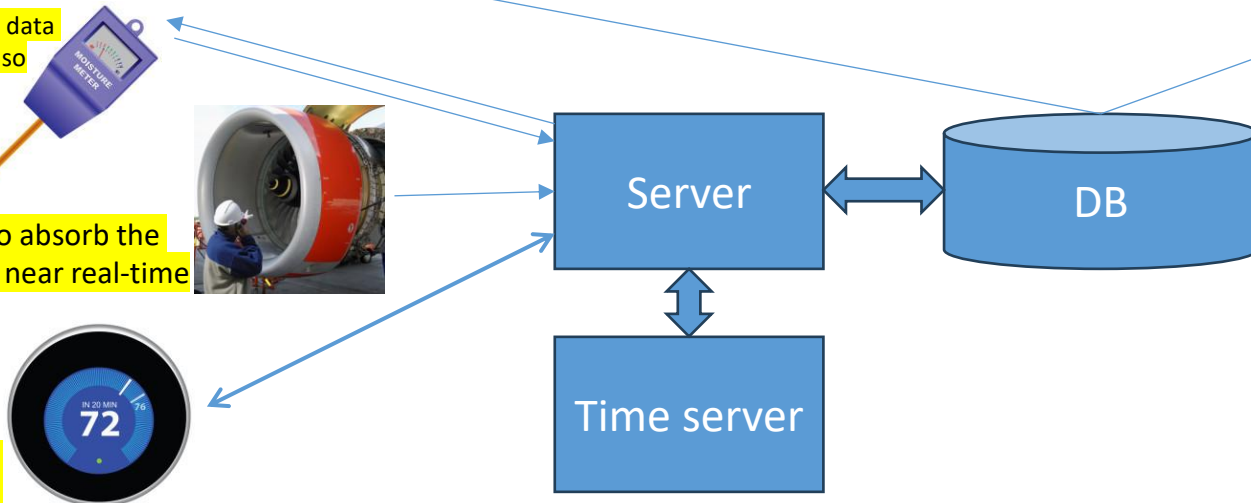
• Passive data

• Active data

• Dynamic data

Need to absorb the data in near real-time

Change the format of the data, the frequency, OTA updates, etc.



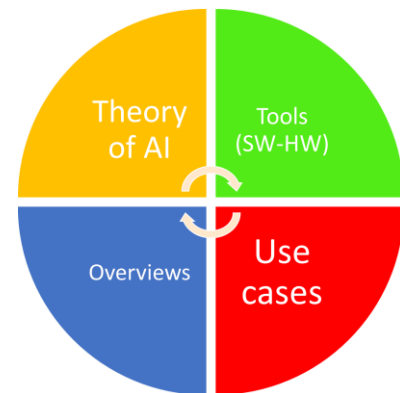
(In the next lessons)



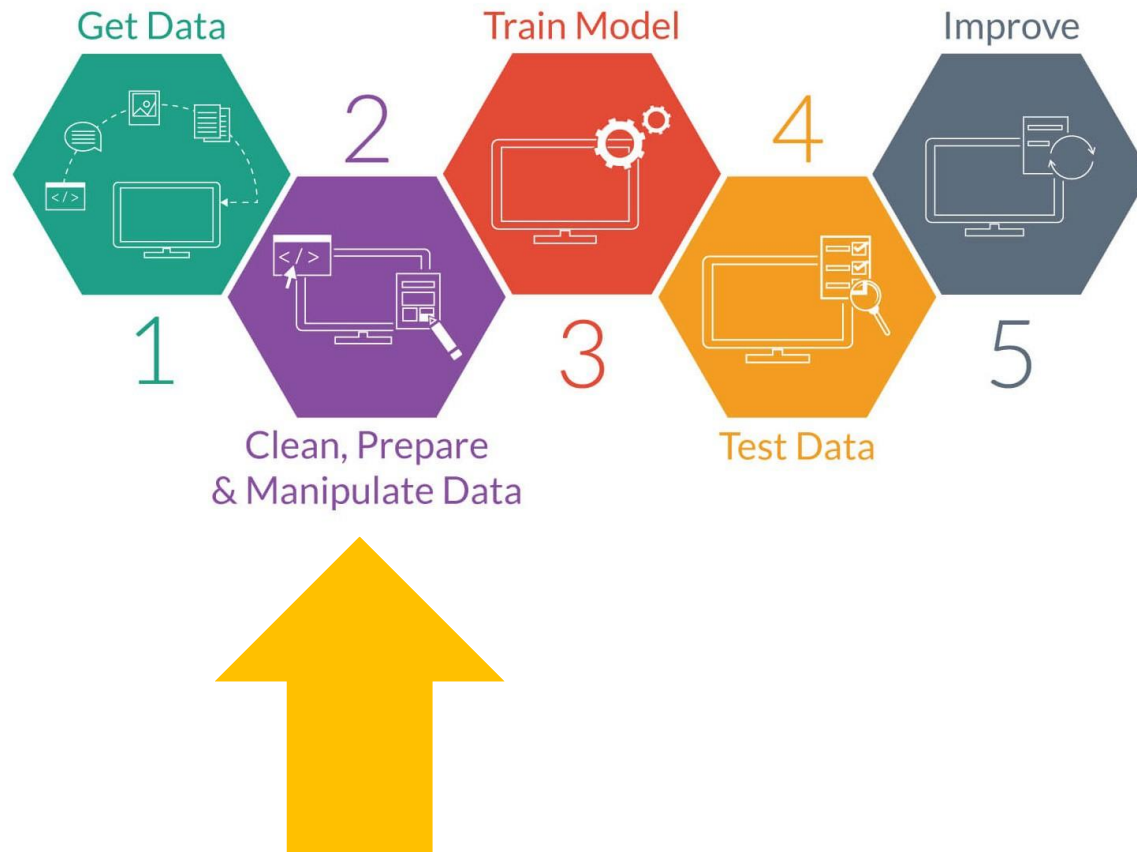
THEORY

Data Preparation

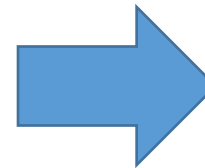
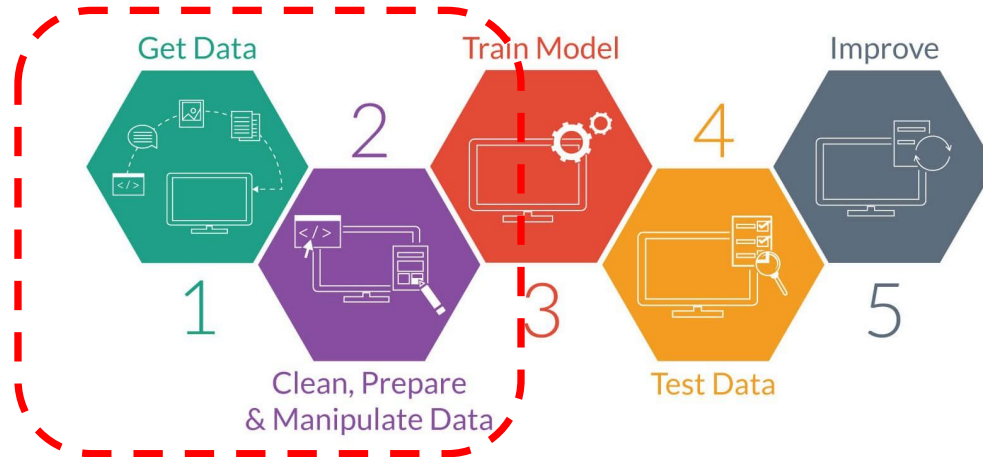
From noisy, messy, unstructured data
to the perfect dataset for your learning phase!



Step 2 of the ML workflow



«Entropy» → Structured data



5.1							
3.5							
1.4							
0.2							

$Y = \text{FUNC}(X)$ that's it!!!

- For every dataset in machine learning or toolbox, is all about to create X and Y

Sample, sample,

- X

Features

5.1								
3.5								
1.4								
0.2								

Class ID, or value to be learnt

- Y

1	0	3	1					
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Data preparation

- A very important part of Data Science.
- It includes two concepts such as *Data Cleaning* and *Feature Engineering*.
- Two **compulsory** steps for achieving better accuracy and performance in the Machine Learning and Deep Learning projects.



Data Preparation

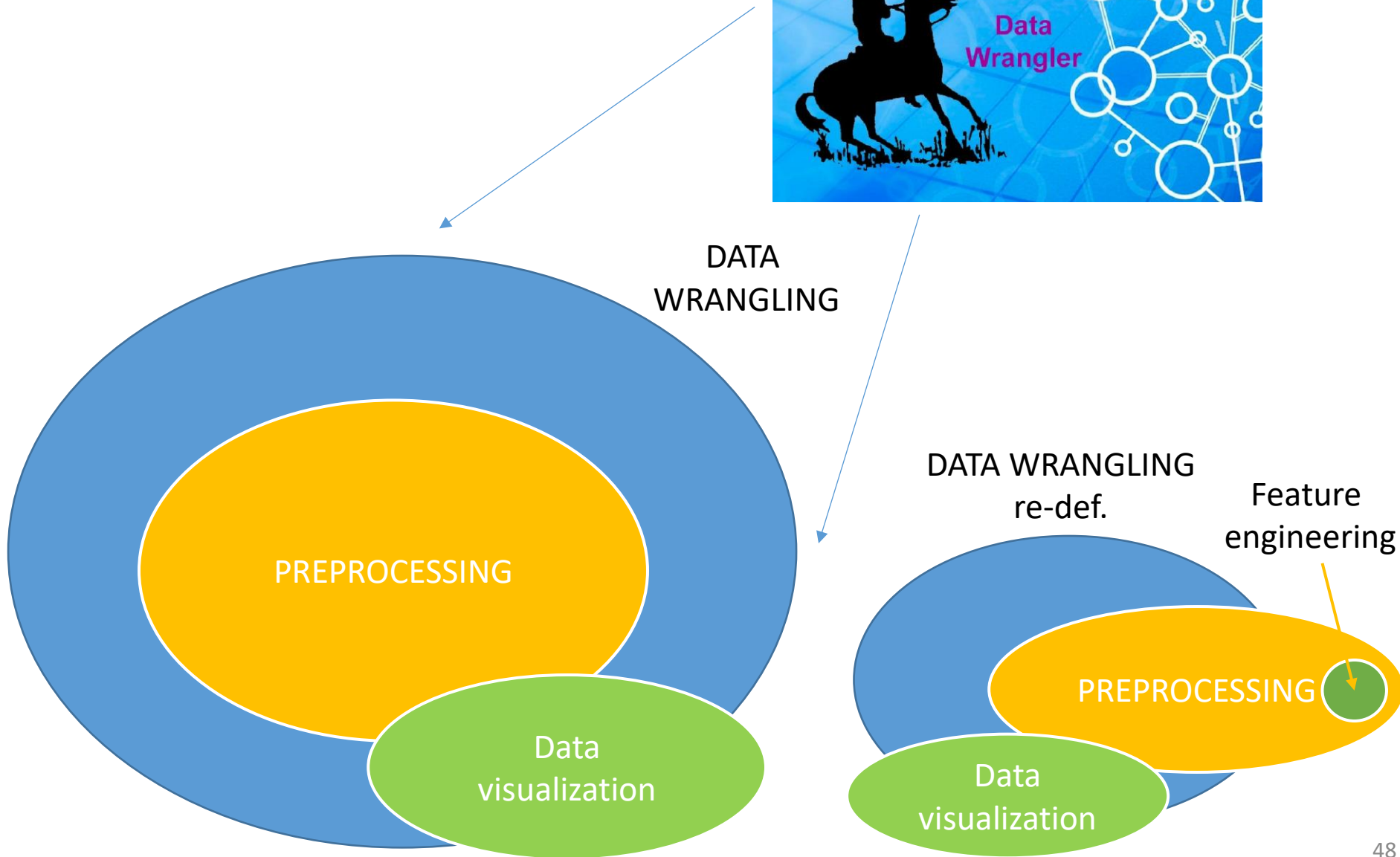


DATA PREPROCESSING



DATA WRANGLING

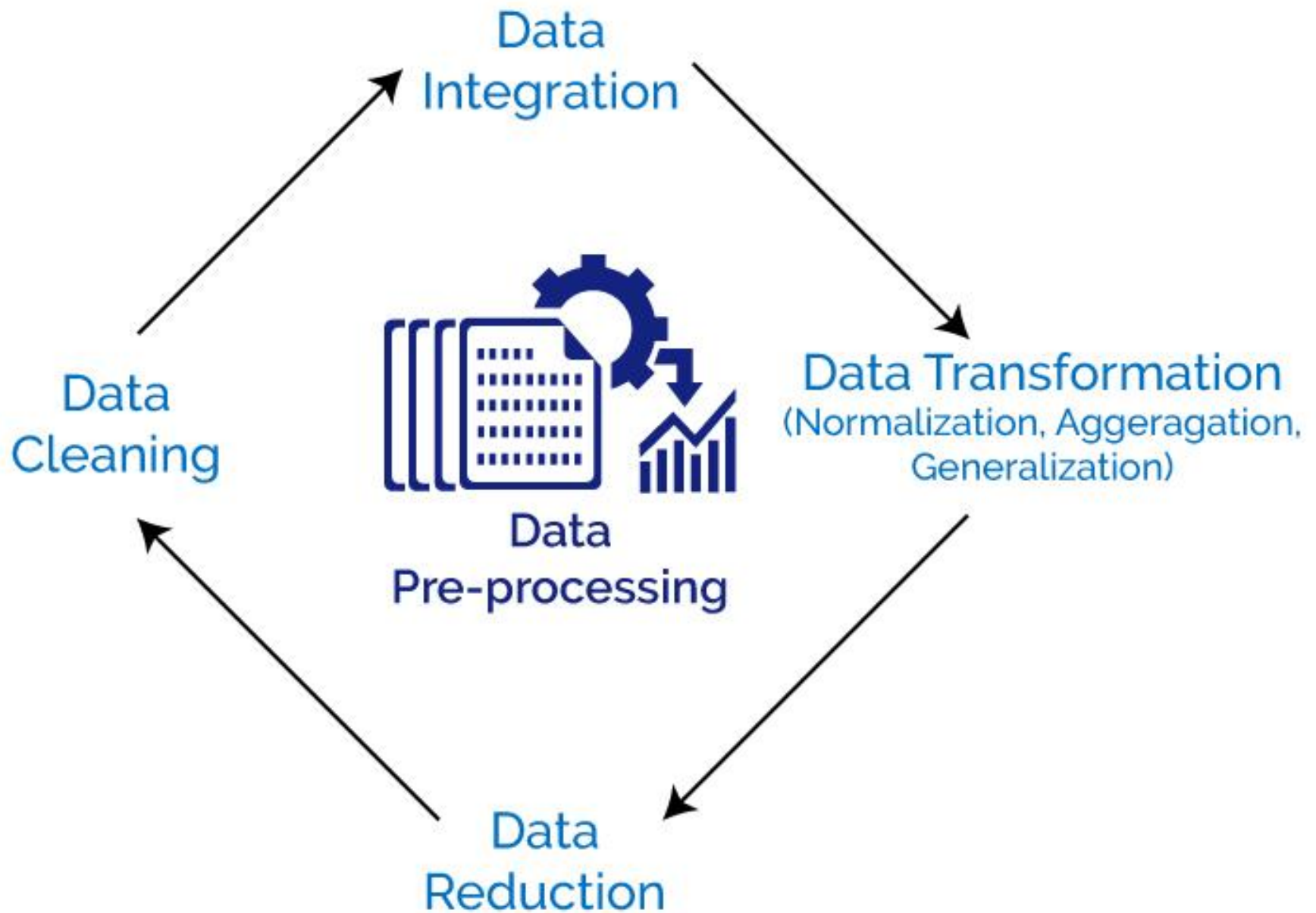
Data Preparation



First things first!

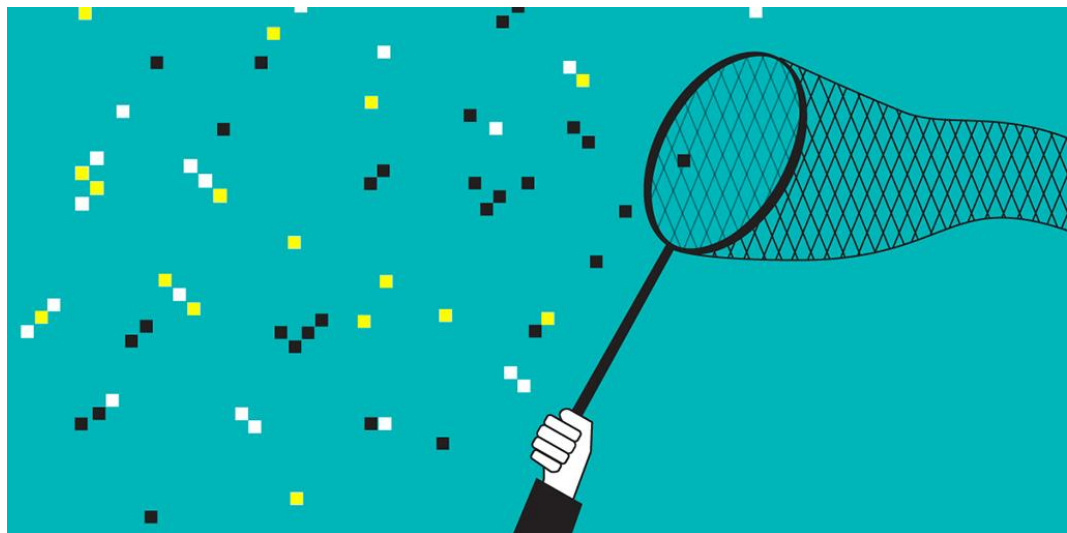
*“It’s an **absolute myth** that you can send an algorithm over **raw data** and have insights pop up” (Jeffrey Heer)*

- The **data wrangling problem** is growing as different types of unstructured data or data in varying formats are pouring in from **sensors, online and from traditional databases**.
- All these data must be cleaned up and organized before data analytics/classifiers/regressors models can be applied.



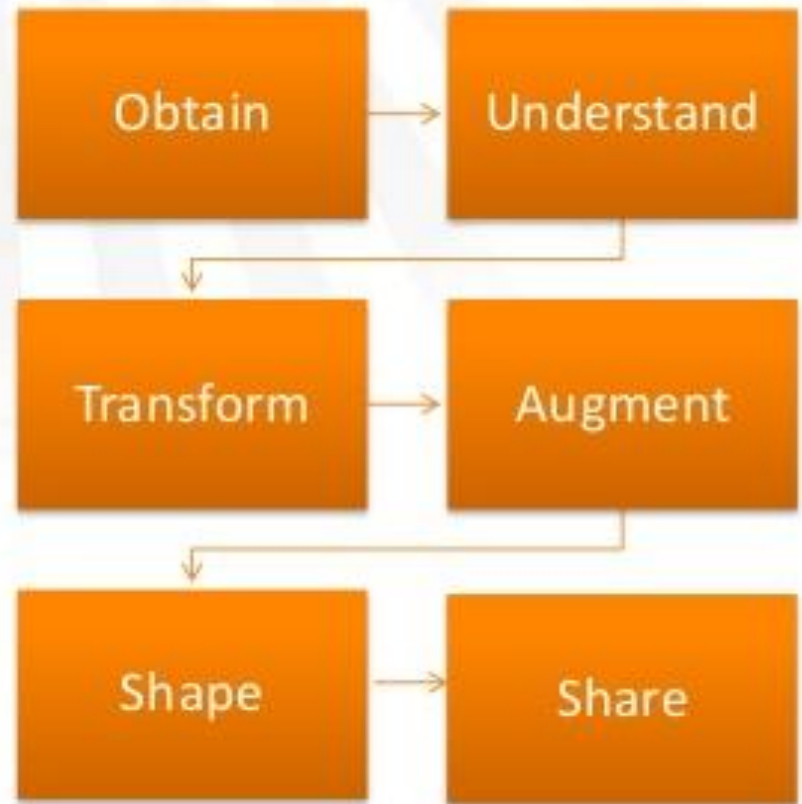


Data wrangling



Data wrangling steps

- Iterative process
- Understand
- Explore
- Transform
- Augment
- Visualize



Data wrangling activities

- Data Preprocessing
- Data Preparation
- Data Cleansing
- Data Scrubbing
- Data Munging
- Data Transformation
- Data Fold, Spindle, Mutilate...



Tasks of Data Wrangling

Discovering

- Firstly, data should be understood thoroughly and examine which approach will best suit.
 - E.g.: if you have a weather data when we analyze the data it is observed that data is from one area and so primary focus is on determining patterns.

Structuring

- As the data is gathered from different sources, the data will be present in various shapes and sizes. Therefore, there is a need for structuring the data in proper format.

Cleaning

- Cleaning or removing of data should be performed that can degrade the performance of analysis.

Enrichment

- Extract new features or data from the given data set to optimize the performance of the applied model.

Validating

- This approach is used for improving the quality of data and consistency rules so that transformations that are applied to the data could be verified.

Data Preprocessing Tools

- **Matlab** (see next slides/lessons)
- **R**: a framework that consists of various packages that can be used for Data Preprocessing like dplyr etc.
- **Weka**: a software that contains a collection of Machine Learning algorithms for Data Mining process. It consists of Data Preprocessing tools that are used before applying Machine Learning algorithms.
- **RapidMiner**: an open-source Predictive Analytics Platform for Data Mining process. It provides the efficient tools for performing exact Data Preprocessing process.
- **Python**: a programming language that provides various libraries that are used for Data Preprocessing.
- **Agents**: Claude Code/Cowork, Codex, Gemini, etc.

Data pre-processing

- is a technique that is used to convert the raw data into a clean data set.
 - Whenever the data is gathered from different sources or it is collected in raw format is not (always) feasible for the analysis.
- Pre-processing includes
 - Data cleaning
 - Data integration
 - Data transformation
 - Data reduction

Data wrangling: Data Preprocessing and Visualization

- Data preprocessing and visualization are one of the first steps to understand the data you have to use
- To find...
 - Outliers
 - Errors
 - Missing values
 - Different scales
 - Distributions
 - Clusters in the dataset
 - Salient features



Data
preparation

Feature
engineering

This is an important
list to be remembered

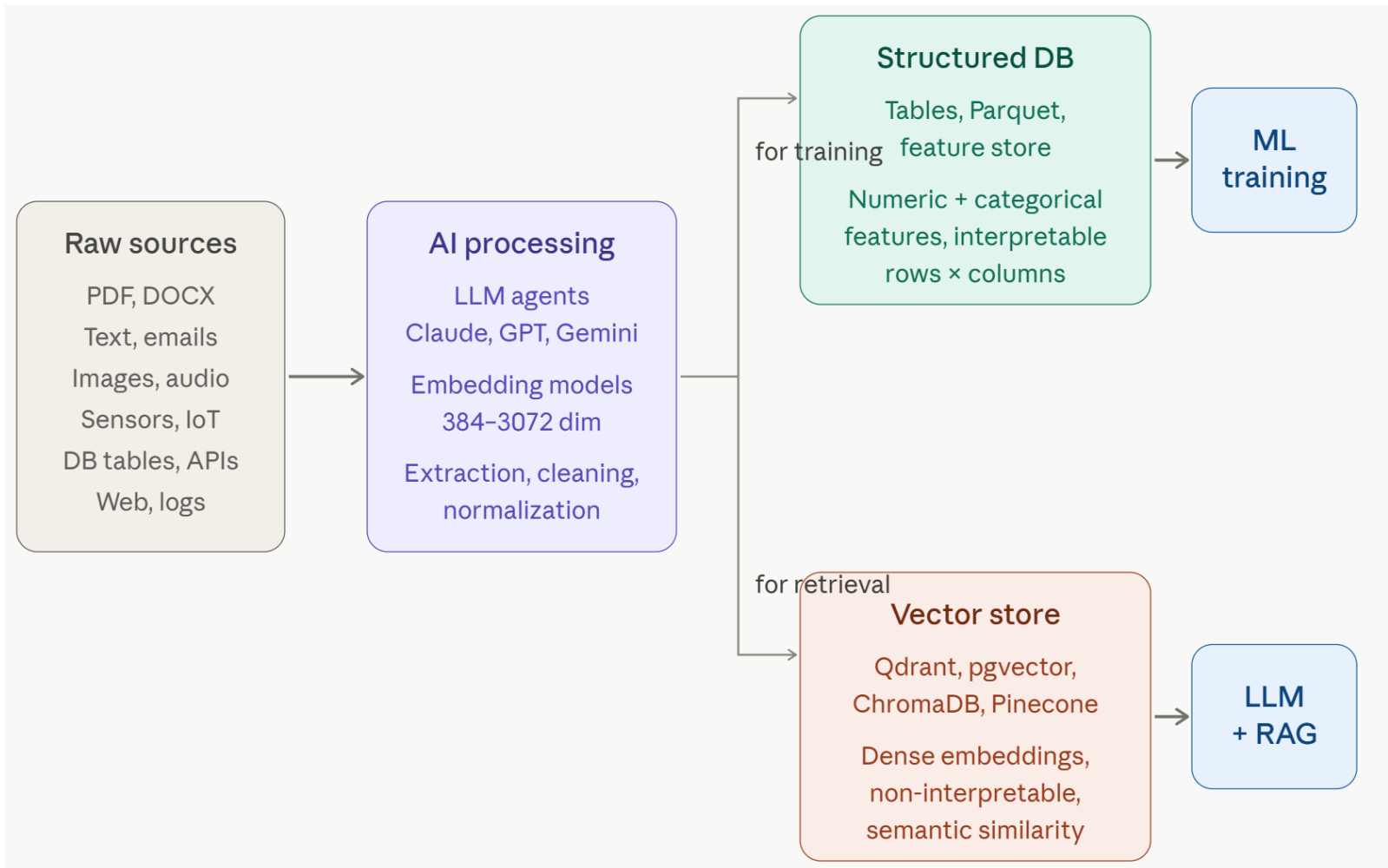
Why is Data Preprocessing is so important?

- Data Preprocessing is necessary because of the presence of unformatted real-world data
 - **Inaccurate data (missing data)**
Many reasons for missing data: data is not continuously collected, a mistake in data entry, technical problems with sensors/transmission/DoS/delayed
 - **The presence of noisy data/erroneous data/outliers**
could be a technological problem of device that gathers data, a human mistake during data entry, etc.
 - **Inconsistent data**
Existence of duplication within data, human data entry, containing mistakes in codes or names, i.e., violation of data constraints and much more

AI agents for data wrangling

- Claude Code, Cowork, Codex CLI, Gemini CLI → autonomous data cleaning, schema inference, entity resolution, regex generation from examples
- Typical use: "given this messy CSV with mixed date formats and units, normalize it" → agent writes + runs + tests the pipeline
- Human role shifts from *writing* the transformation to *validating* it

AI agents for data wrangling and DB creation



LLMs for structured extraction from unstructured data

- Input: free text (reports, emails, clinical notes, PDFs)
→
Output: typed JSON records / MD
- Replaces most NER (Named Entity Recognition) + rule-based parsing pipelines
- Example: 1000 PDF lab reports → DataFrame with 15 normalized columns in one prompt

Named Entity Recognition (NER)

Classical NLP technique for extracting named entities from free text: people, locations, dates, quantities, organizations, medical codes, etc

- **Input:** unstructured text "Patient Mario Rossi, 54 years old, was admitted on 03/12/2026 to San Raffaele Hospital with a diagnosis of acute myocardial infarction, ICD-10 code I21.9"
- **Output:** labeled entities
 - PERSON: Mario Rossi
 - AGE: 54
 - DATE: 03/12/2026
 - ORG: San Raffaele Hospital
 - DIAGNOSIS: acute myocardial infarction
 - ICD10: I21.9

Vector databases

the new storage layer for unstructured data

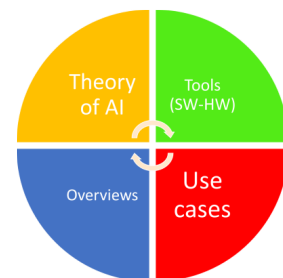
- Qdrant, Pinecone, pgvector, ChromaDB, Weaviate, Milvus
- Store embeddings (dense vectors, typically 384–3072 dim) instead of raw text/images
- Enable semantic search, RAG, deduplication by meaning (not by string match)
- Typical pipeline:
raw doc → embedding model → vector DB → similarity query

Caveat!

GIGO still applies



- Agents hallucinate schemas; LLM extractions drift; embeddings reflect training biases
- Always validate: type checks, range checks, cross-source reconciliation, human spot-check on samples
- Rule: **AI-assisted $\neq$ AI-trusted**
- **Lesson learned:**
Data preparation in 2026 is still Step 2 of the ML workflow — but Step 2 itself now uses AI



THEORY

Missing Data

Dealing with data complexity



Missing data representation

	col1	col2	col3	col4	col5
0	2	5.0	3.0	6	NaN
1	9	NaN	9.0	0	7.0
2	19	17.0	NaN	9	NaN

NaN

Member of a numeric data type that can be interpreted as a value that is undefined, unrepresentable, or missing (encoded with the [IEEE 754](#) standard)

Note: NaN is defined by IEEE 754 only for floating-point types — integers cannot hold NaN, which is why modern libraries (pandas Int64, Arrow, Polars) introduced a type-agnostic NA marker for nullable columns.

```
data = {"sensor_id": [1, 2, 3, 4], "rpm": [2400, 2401, None, 2403]}
```

Missing data example

3 Hospitals are merging data of their patients over blood test screening exams
But the labs have different systems and instruments with a different number of features

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1		Hospital 1				Hospital 2				Hospital 3				
2		PZ1	PZ2	PZ3	PZ4	PZ5	PZ6	PZ7	PZ8	PZ9	PZ10	PZ11	PZ12	PZ13
3	Feat. 1	0,744422	0,205193	0,766586	0,487325	0,775158	0,634885	0,765383	0,269233	0,301776	0,366692	0,83481	0,048661	0,068629
4	Feat. 2	0,341608	0,598078	0,431988	0,386859	0,311953	0,595674	0,914989	0,437615	0,264475	0,660899	0,230343	0,342688	0,181113
5	Feat. 3	0,406022	0,950997	0,045678	0,994392	0,848554	0,221909	0,034978	0,761679	0,763025	0,912281	0,165851	0,008296	0,016205
6	Feat. 4	0,456926	0,94847	0,094555	0,310251	0,503689	0,641912	0,380669	0,828563	0,446763	0,084337	0,057833	0,355379	0,738135
7	Feat. 5	0,380072	0,351067	0,658412	0,837005	0,019425	0,686687	0,191358	0,832462	0,833955	0,010785	0,608542	0,537297	0,845168
8	Feat. 6	0,662976	0,856301	0,696342	0,688191	0,552233	0,150594	0,517138	0,903131	0,717526	0,158101	0,57893	0,178603	0,291695
9	Feat. 7	0,25033	0,335838	0,395532	0,544964	0,198371	0,629432	0,843826	0,702729	0,962303	0,405384	0,965353	0,093214	0,439128
10	Feat. 8	0,456787	0,54951	0,217909	0,844122	0,527656	0,92943	0,18003	0,644716	0,408152	0,100464	0,372383	0,718214	0,901732
11	Feat. 9					0,929722	0,369789	0,170243	0,226735	0,575054	0,11046	0,513469	0,286501	0,696153
12	Feat. 10					0,116038	0,808077	0,579666	0,096338	0,588189	0,553363	0,745312	0,123346	0,028077
13	Feat. 11					0,793393	0,630495	0,406624	0,525547					
14	Feat. 12					0,253424	0,199476	0,47691	0,898241					

Data Harmonization is crucial here since the sources of data can be different!

Es: mg/dL → mmol/L → etc. etc.

Remove incomplete features

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1		Hospital 1				Hospital 2				Hospital 3				
2		PZ1	PZ2	PZ3	PZ4	PZ5	PZ6	PZ7	PZ8	PZ9	PZ10	PZ11	PZ12	PZ13
3	Feat. 1	0,744422	0,205193	0,766586	0,487325	0,775158	0,634885	0,765383	0,269233	0,301776	0,366692	0,83481	0,048661	0,068629
4	Feat. 2	0,341608	0,598078	0,431988	0,386859	0,311953	0,595674	0,914989	0,437615	0,264475	0,660899	0,230343	0,342688	0,181113
5	Feat. 3	0,406022	0,950997	0,045678	0,994392	0,848554	0,221909	0,034978	0,761679	0,763025	0,912281	0,165851	0,008296	0,016205
6	Feat. 4	0,456926	0,94847	0,094555	0,310251	0,503689	0,641912	0,380669	0,828563	0,446763	0,084337	0,057833	0,355379	0,738135
7	Feat. 5	0,380072	0,351067	0,658412	0,837005	0,019425	0,686687	0,191358	0,832462	0,833955	0,010785	0,608542	0,537297	0,845168
8	Feat. 6	0,662976	0,856301	0,696342	0,688191	0,552233	0,150594	0,517138	0,903131	0,717526	0,158101	0,57893	0,178603	0,291695
9	Feat. 7	0,25033	0,335838	0,395532	0,544964	0,198371	0,629432	0,843826	0,702729	0,962303	0,405384	0,965353	0,093214	0,439128
10	Feat. 8	0,456787	0,54951	0,217909	0,844122	0,527656	0,92943	0,18003	0,644716	0,408152	0,100464	0,372383	0,718214	0,901732
11	Feat. 9					0,929722	0,369789	0,170243	0,226735	0,575054	0,11046	0,513469	0,286501	0,696153
12	Feat. 10					0,116038	0,808077	0,579666	0,096338	0,588189	0,553363	0,745312	0,123346	0,028077
13	Feat. 11					0,793393	0,631415	0,405624	0,525577					
14	Feat. 12					0,253424	0,199476	0,477691	0,658241					

What if

Feature #12 is the feature you are looking for to detect the studied cancer?

Removing incomplete vectors

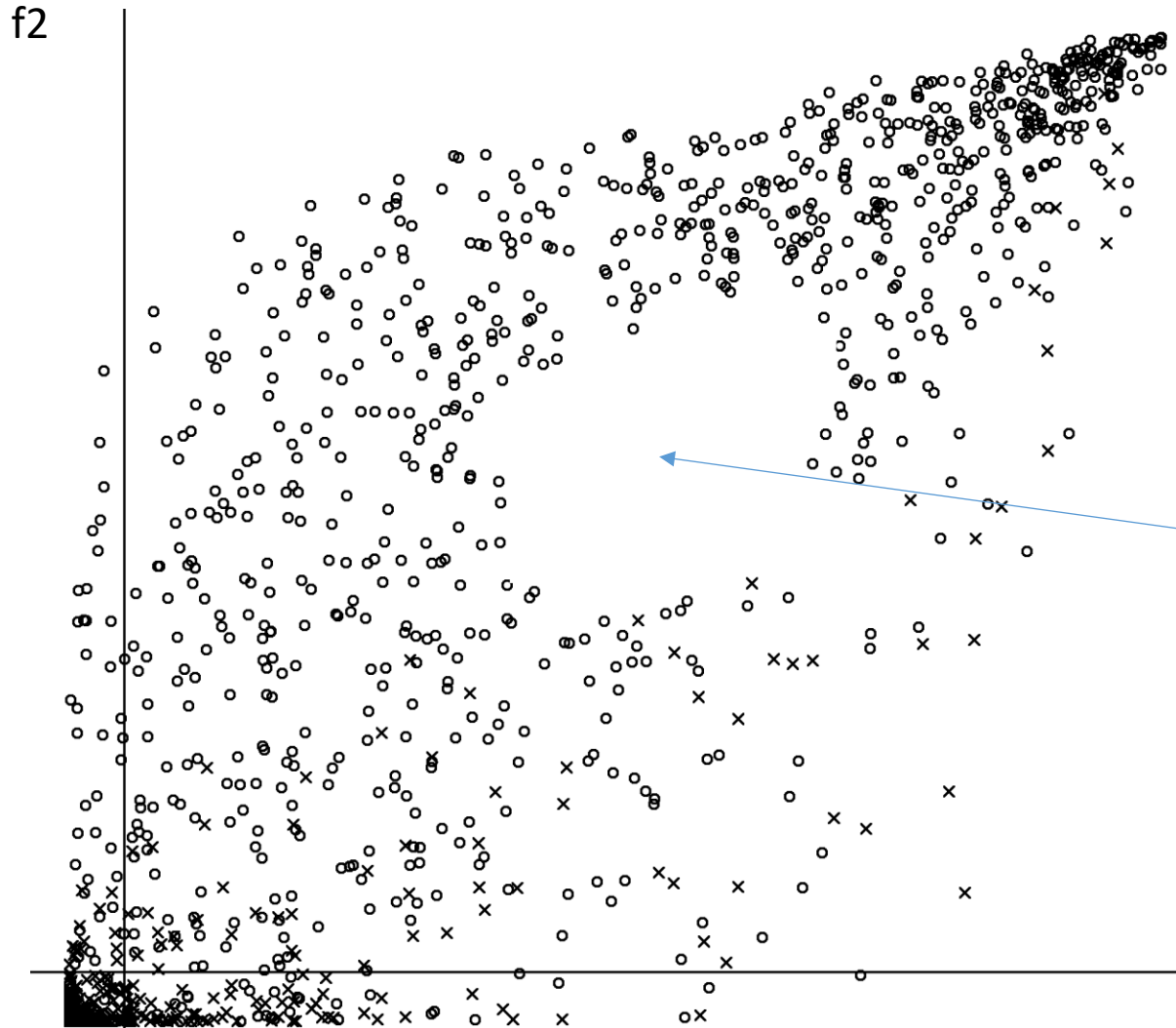
	A	B	C	D	E	F	G
1	Original Data set				Data set after Listwise deletion		
2	Name	Age	Gender		Name	Age	Gender
3	Robin	28	Male		Robin	28	Male
4	Heather	29	Female		Heather	29	Female
5	Jamie	22			Carl	32	Male
6	Carl	32	Male		Sarah	26	Female
7		35	Male				
8	Sarah	26	Female				
9							
10							

Bye-bye Jamie and Mr “35 Male” → from 6 records to only (complete) 4

Is this loss of information acceptable for your application?

A «topological» miss

The matrix of data is complete, but
the feature space is
not uniformly covered



x = Reject
o = Good

????

Without proper
data visualization
is hard to find this
problem....

f1

Data Preprocessing: missing data (1)



Ignoring the missing record

It is the simplest and efficient method for handling the missing data.

- How your learning algorithm is managing them?!
- This method should not be performed at the time
 - when the number of missing values are immense
 - or when the missing data problem can be solved (debugging/re-designing the experiment) and not just ignoring the problem causing the missing data.

	0	0
0	25	35000
1	27	40000
2	50	54000
3	35	nan
4	40	60000
5	35	58000
6	nan	52000
7	48	79000
8	50	83000
9	37	nan
10	21	24000
11	nan	60000
12	63	70000

Data Preprocessing: missing data (2)



Filling the missing values manually

- This is **one of the best-chosen** methods.
- But there is one limitation that when there are **large data set**, and **missing values are significant** then, this approach is not efficient as it becomes a time-consuming task.



For example, in medical trials
every single datum
is very important and expensive.



Excluding a vector
is not a good first strategy!

I'm sorry, I have some missing values from the sensors... Can we do that again?



Data Preprocessing: missing data (3)



Filling using computed values

- The missing values can also be occupied by **computing mean, mode or median** of the observed given values.
- You can also find **the most similar column** and use it to copy the missing values (ex. using kNN classifiers)
- Generate values by using any ML or Deep Learning algorithm
 - Drawback: **it can generate bias within the data.**

A basic use case:

using a linear model

```
x = [-4*pi:0.1:0, 0.1:0.2:4*pi];  
A = sin(x);
```

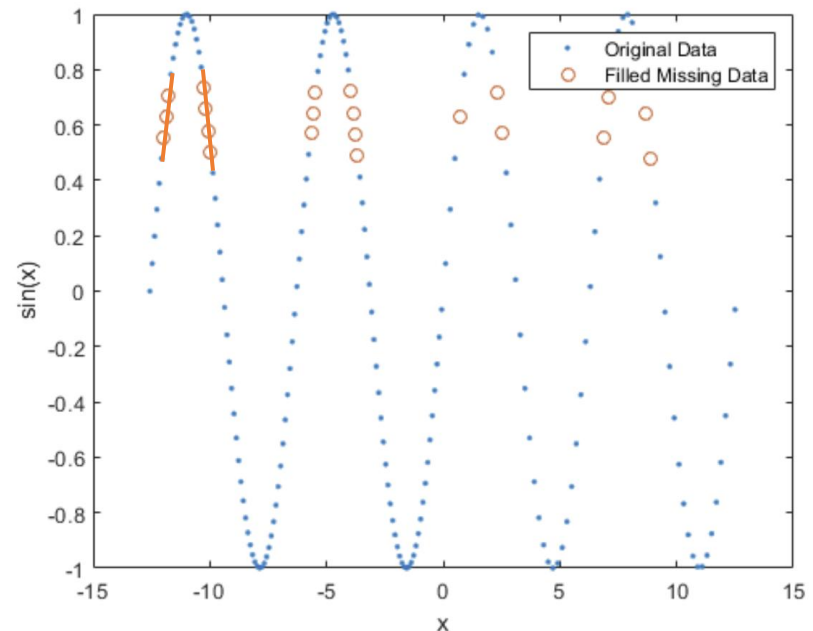
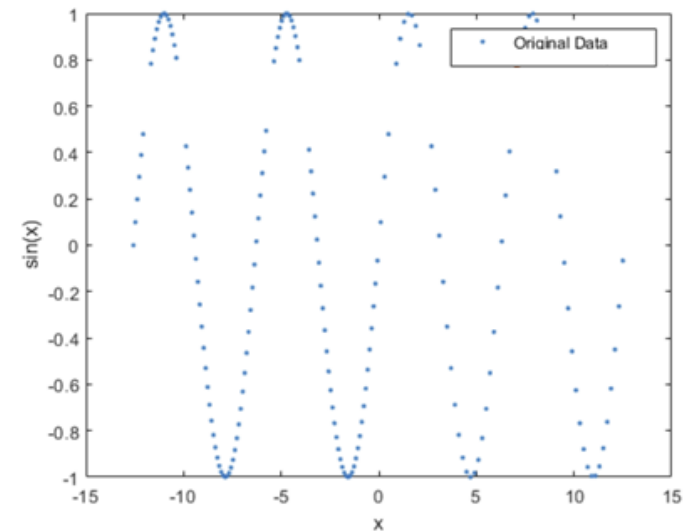
```
A(A < 0.75 & A > 0.5) = NaN;
```

- F: Fill the missing data using linear interpolation
TF: corresponds to the values of F that were filled

```
[F,TF] = fillmissing(A,'linear','SamplePoints',x);
```

Plot the original data and filled data.

```
plot(x,A,'.', x(TF),F(TF),'o')  
xlabel('x');  
ylabel('sin(x)')  
legend('Original Data','Filled Missing Data')
```



The choice of the model to be used form interpolate the missing points will affect the learning of subsequent model

Missing data:

Possibile solutions in brief

- STEP 0: Quantify the problem (Row? Cols?)
- **Do Nothing**
 - let the algorithm handle the missing data
→ ok, but what is it doing? Explainability and cross validation can be compromised. Some method will ignore, or crash, erratic, ...
- Filling the missing values **manually** (if possible and ask the expert about data distribution and ranges)
- **Statistic** appr.: Mean, Mode, Median, Tabu-value (☹!) or constant
 - WARNING!! In case of categorical features → real values
- K-Nearest Neighbor/s: **find the most similar vector**/vectors in input and copy the missing values

In brief...

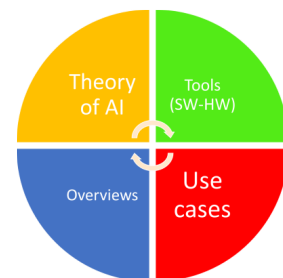




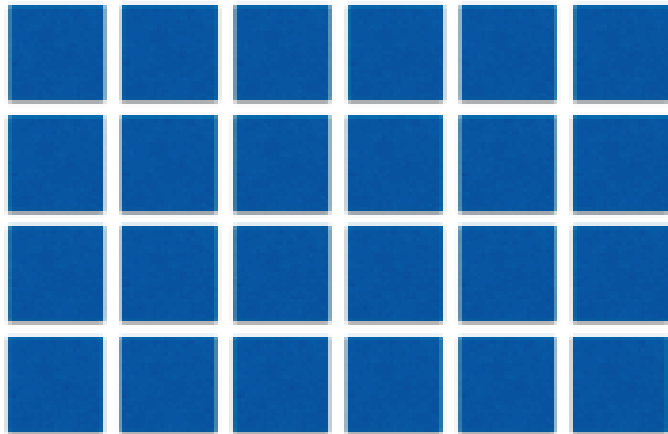
THEORY

Structured and unstructured Data

Dealing with data complexity

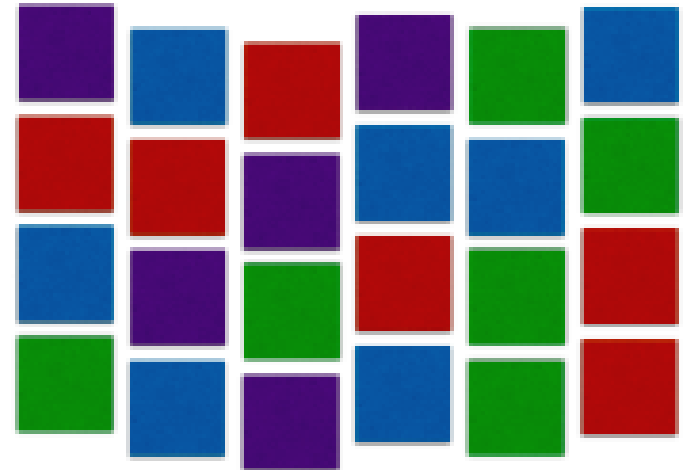


Structured Data



What you find in a DB
(typically)

Unstructured Data



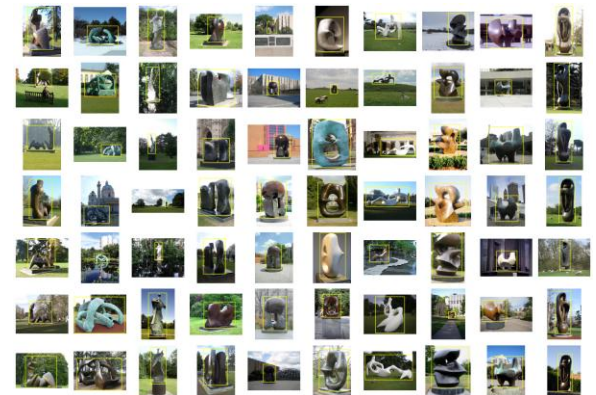
What you find in the 'wild'
(text, images, audio, video)

Structured data (Good!)

- Structured data usually resides in relational databases (RDBMS).
 - Fields store length-delineated data phone numbers, Social Security numbers, or ZIP codes.
 - Even text strings of variable length like names are contained in records, making it a simple matter to search.
- Data may be human- or machine-generated as long as the data is created within an RDBMS structure.
- This format is eminently searchable both with human generated queries and via algorithms using type of data and field names, such as alphabetical or numeric, currency or date.

Unstructured Data (Bad...)

- Unstructured data is essentially everything else. Unstructured data has internal structure but is not structured via pre-defined data models or schema.
 - It may be textual or non-textual, and human- or machine-generated. It may also be stored within a non-relational database like NoSQL



Semi-structured

UNSTRUCTURED



SEMI-STRUCTURED



STRUCTURED



Parquet



Apache
orc

More flexible

More efficient storage and performance



Typical human-generated unstructured data includes:

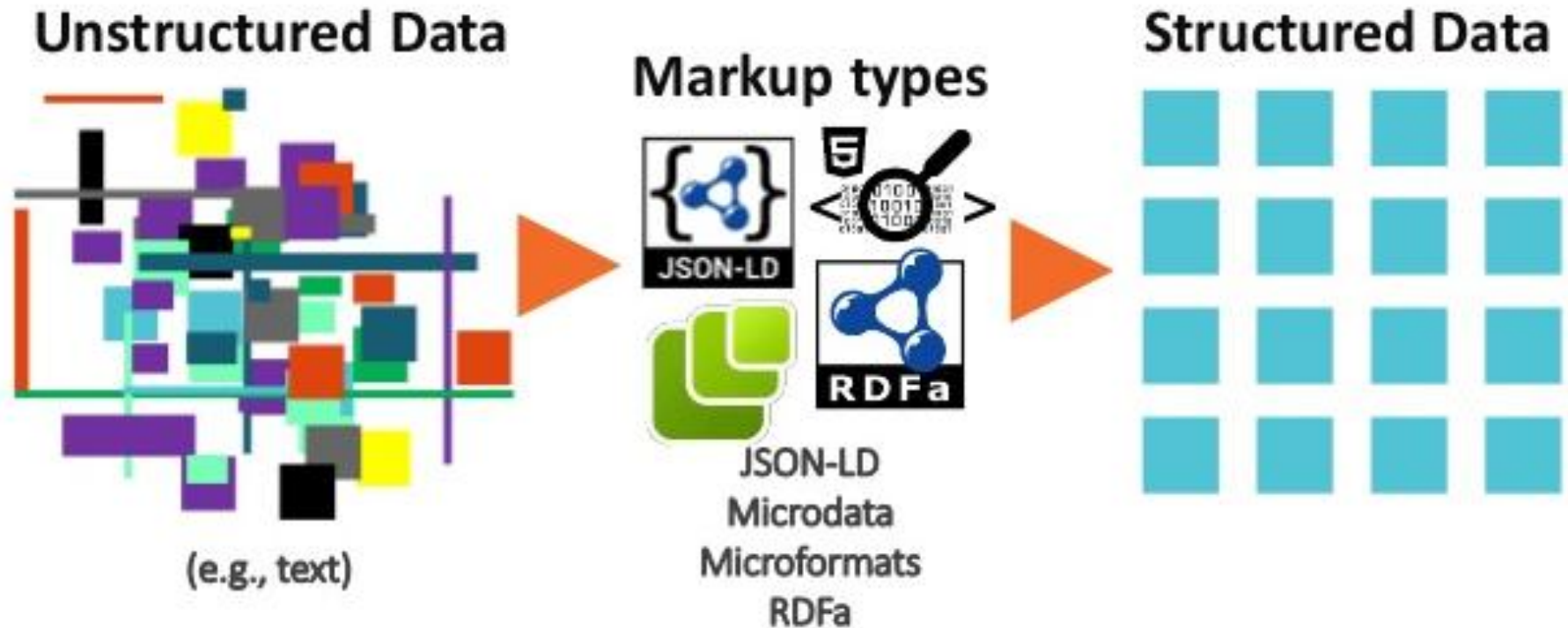
- **Text files:** Word processing, spreadsheets, presentations, email, logs.
- **Email:** Email has some internal structure thanks to its metadata, and we sometimes refer to it as [semi-structured](#). However, its message field is unstructured and traditional [analytics tools](#) cannot parse it.
- **Social Media:** Data from Facebook, Twitter, LinkedIn.
- **Website:** YouTube, Instagram, photo sharing sites.
- **Mobile data:** Text messages, locations.
- **Communications:** Chat, IM, phone recordings, collaboration software.
- **Media:** MP3, digital photos, audio and video files.
- **Business applications:** MS Office documents, productivity applications.



Typical machine-generated unstructured data includes:

- **Satellite imagery:** Weather data, land forms, military movements.
- **Scientific data:** Oil and gas exploration, space exploration, seismic imagery, atmospheric data.
- **Digital surveillance:** Surveillance photos and video.
- **Sensor data:** Traffic, weather, oceanographic sensors.

A web example of data transformation

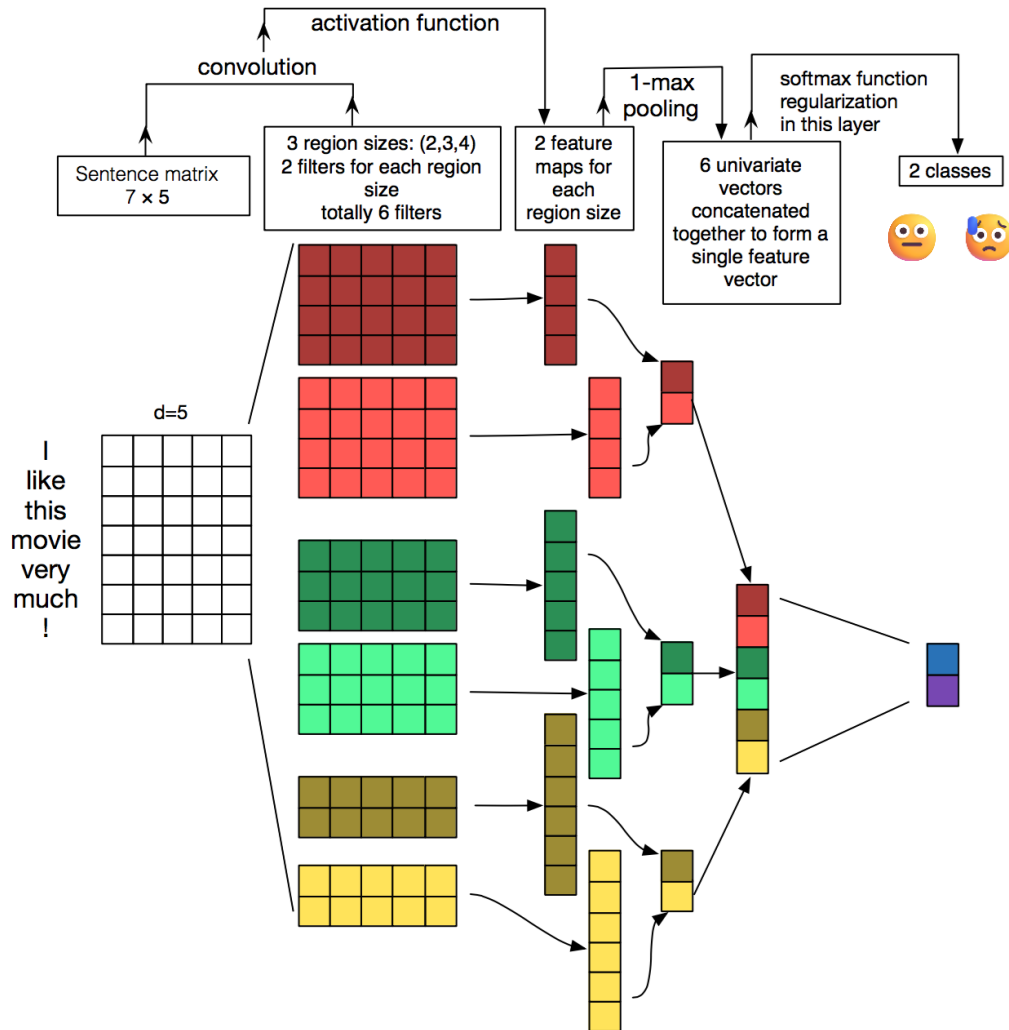


Resource Description Framework in Attributes (RDFa) is a W3C Recommendation that adds a set of attribute-level extensions to HTML, XHTML and various XML-based document types for embedding rich metadata within Web documents.

Neural networks and unstructured data



It is not strictly compulsory to have structured data to use ML



Some examples of text classification are:

- Understanding audience sentiment (😄 😐 😞) from social media
- Detection of spam & non-spam emails
- Auto tagging of customer queries
- Categorization of news articles into predefined topics

Main points



- Using datasets without previous activities is useless dangerous and it wastes your time
- Study, cleaning, harmonize... and comprehend basic relationship is mandatory
- Dealing with missing data
- Structured and unstructured data